Posets And Lattices

A) Posets

I) Definition

A Relation R on a set P is called a Partial Order Relation if it satisfies the following conditions →

1) Reflexive → for any $a \in P$, aRa
2. Antisymmetric → If aRb and bRa then $a = b$
3. Transitive → If aRb and bRc then aRc

Notation $(P, \leq)$, $(P, \preccurlyeq)$

*) The set P together with a partial order relation is called a partially ordered set or Poset

If $a \preccurlyeq b$, read as a precedes b

Q Check whether the set $(R, \leq)$ is a Poset.

Yes

Q Check whether $(Z, >)$ is a Poset.

No.

(II) Diagrammatic Representation of POSET

- It is called Hasse Diagram
- Let $(P, <)$ be a Poset and $x, y \in P$

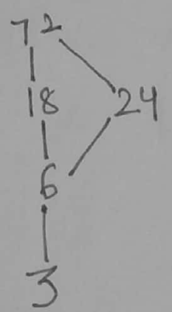
The element y is called cover of x if $x < y$ and no $z \in P$ exists such that $x < z < y$

i.e y is called cover of x, if y is the immediate successor of x.

Eg

$P = \{3, 6, 18, 24, 72\}$

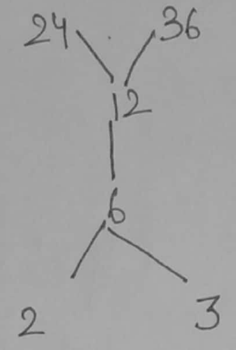
$(P, \leq) \rightarrow$ a factor of b



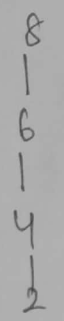
→ Elements below a point follow the property but elements at the same level or above do not follow the property.

Q Let $x = \{2, 3, 6, 12, 24, 36\}$ and the relation be such that $x \leq y$ if x divides y.

Draw the Hasse Diagram of $(x, \leq)$



A) Let $P = \{2, 4, 6, 8\}$ $(P, \leq) \rightarrow$ a less than b



<u>Observation</u>

→ All elements in a sequence without branches.

→ It forms a chain

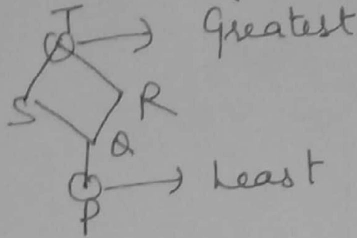
Here every 2 elements are comparable or satisfy the property

This is called a Totally Ordered Set or (TOSET

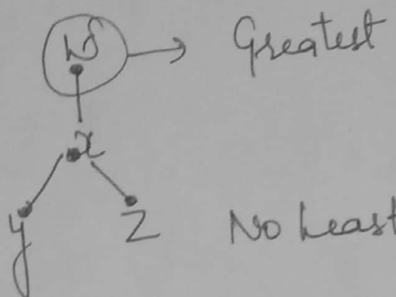
(III) Elements of POSET

Let $\langle P, \preceq \rangle$ be a Poset

(i) **Least Element** $\{ y \in P \mid \forall x \in P, y \preceq x \}$



(ii) **Greatest Element** $\{ y \in P \mid \forall x \in P, x \preceq y \}$

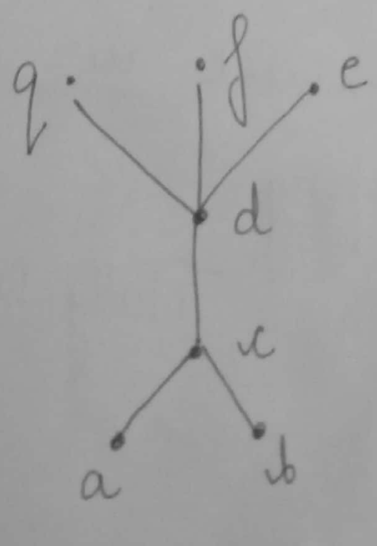


No Least Element

(*) Greatest and Least Element are unique, if they exist.

(iii) **Minimal Element** $\{ y \in P \mid \text{No } x \in P, x < y \}$

(iv) **Maximal Element** $\{ y \in P \mid \text{no } x \in P, y < x \}$



maximal $\{g, f, e\}$

minimal $\{a, b\}$

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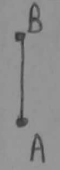
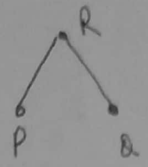
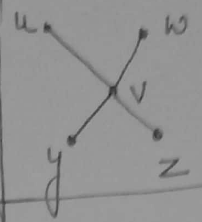
*) Maximal & Minimal Element need not be Unique.

*) Distinct minimal / maximal members are incomparable.

→

			
<u>Least</u>	A	—	—
<u>Greatest</u>	B	R	—
<u>Minimal</u>	$\{A\}$	$\{P, Q\}$	$\{y, z\}$
Maximal	$\{B\}$	$\{R\}$	$\{u, w\}$

(*) <u>Lower and Upper Bound</u>

Let $\langle P, \leq \rangle$ be the Poset and $\langle Q, \leq \rangle$ be a Subset.

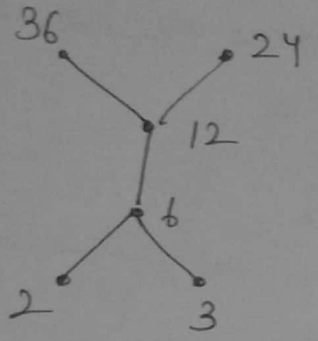
$\langle P, \leq \rangle \to$ Subset $\langle Q, \leq \rangle$

$y \in P$ $\qquad$ $x \in Q$

(v) <u>Lower Bound</u> $\quad \forall x, \; y \leq x$

(vi) <u>Upper Bound</u> $\quad \forall x, \; x \leq y$

Eg $\langle H, \preceq \rangle$

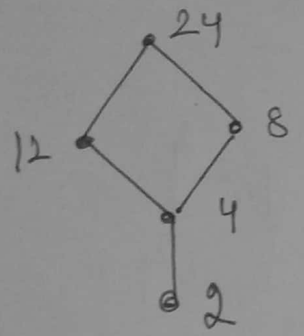


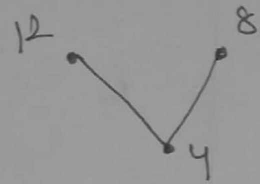
$\langle N, \preceq \rangle$

LB : $\{2, 3, 6\}$

UB : $\{36, 24, 12\}$

Eg.





LB : $\{2, 4\}$

UB : $\{24\}$

*) Boundary element included

*) Also Least in LB and Greatest in UB

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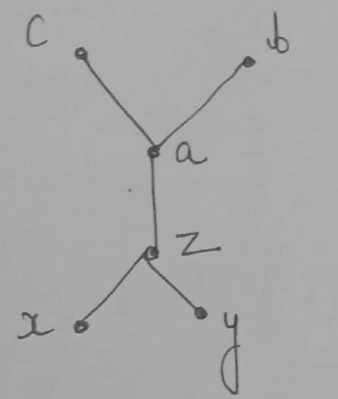
(vii) <u>Greatest Lower Bound (GLB) or Infimum</u> :

If the set of lower bounds of Q has a greatest element, then it is called GLB of Q.

(viii) If the set of upper bounds of Q has a least element, then it is called the <u>least upper bound (LUB) or Supremum</u>

Eg.



$\langle P, \leq \rangle$

$\langle Q, \leq \rangle$

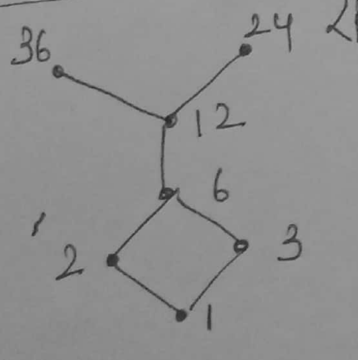
$LB = \{x, y, z\} \rightarrow GLB$

$UB = \{c, b, a\} \rightarrow LUB$

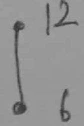
*) If GLB and LUB exist, then they are Unique

*) The elements in LB and UB closest to the subset element is GLB or LUB.

Eg.



$\langle P, \leq \rangle$



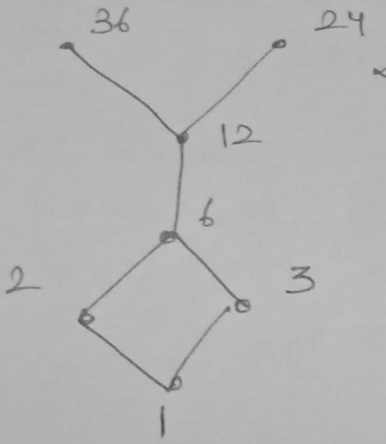
$\langle Q, \leq \rangle$

$LB = \{1, 2, 3, 6\} \rightarrow GLB$

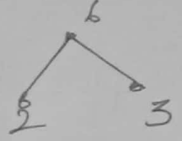
$UB = \{12, 24, 36\}$

$12 \downarrow$ LUB

Eg 7)



$\langle P, \leq \rangle$



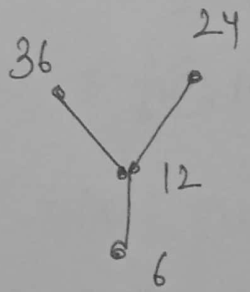
$\langle R, \leq \rangle$

$LB = \{1\} \rightarrow$ GLB

$UB = \{6, 12, 24, 36\}$

$\downarrow$ LUB

Eg



$\langle S, \leq \rangle$

$LB = \{1, 2, 3, 6\} \rightarrow$ GLB

$UB = \{\ \} \rightarrow$ No LUB

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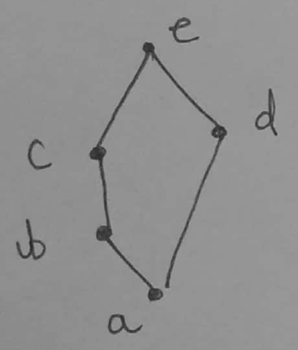
(B) **Lattices** → A Poset $(L, \leq)$ is called a Lattice if every pair of elements in L has an LUB and a GLB.

→ The GLB of x and y is called the meet of x and y, denoted as $x \wedge y$ or $x \cdot y$

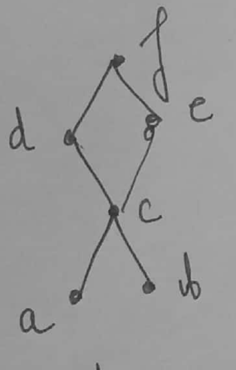
→ The LUB of x and y is called the join of x and y and is denoted as $x \vee y$ or $x + y$

→ Every Chain is a lattice.

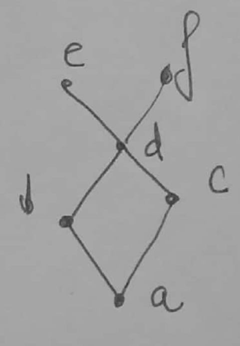
Eg.



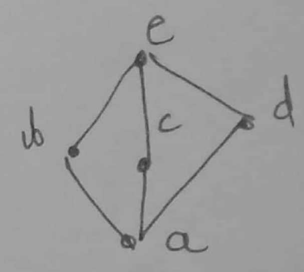
(a)



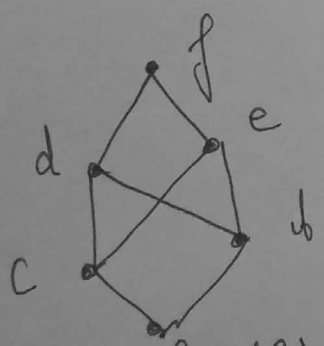
(b)



(c)



(d)



(e)

→ (a) and (d) are Lattices

→ (b) is Not a Lattice because $a \wedge b$ does not exists.

→ (c) is Not a Lattice because $e \vee f$ does not exists.

→ (e) is Not a Lattice because $d \wedge e$ as well as $b \vee c$ does not exist.